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The Effect of Temperature Variability on Biological Responses of Ectothermic Animals—A Meta-Analysis

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ABSTRACT

Climate change is altering temperature means and variation, and both need to be considered in predictions underpinning conservation. However, there is no consensus in the literature regarding the effects of temperature fluctuations on biological functions. Fluctuations may affect biological responses because of inequalities from non-linear responses, endocrine regulation or exposure to damaging temperatures. Here we establish the current state of knowledge of how temperature fluctuations impact biological responses within individuals and populations compared to constant temperatures with the same mean. We conducted a meta-analysis of 143 studies on ectothermic animals (1492 effect sizes, 118 species). In this study, 89% of effect sizes were derived from diel cycles, but there were no significant differences between diel cycles and shorter (<8 h) or longer (>48 h) cycles in their effect on biological responses. We show that temperature fluctuations have little effect overall on trait mean and variance. Nonetheless, temperature fluctuations can be stressful: fluctuations increased 'gene expression' in aquatic animals, which was driven mainly by increased hsp70. Fluctuating temperatures also decreased longevity, and increased amplitudes had negative effects on population responses in aquatic organisms. We conclude that mean temperatures and extreme events such as heat waves are important to consider, but regular (particularly diel) temperature fluctuations are less so.

1 | Introduction

The thermal environment influences most biological and ecological processes (Kingsolver and Buckley 2017). Anthropogenic activities are increasing the magnitude and frequency of extreme climatic events (Lee 2023), leading to changes in both mean temperature and amplitude of temperature fluctuations (Angélil et al. 2017; Qian and Zhang 2015; Vázquez et al. 2017). In natural environments, these changes can manifest as unpredictable thermal variability at different timescales (Mora et al. 2013; Wedler, Pinto, and Hochman 2023). Ectotherms are particularly susceptible to thermal variability because their internal body temperature is directly influenced by their surrounding environment (Chown et al. 2010; Huey and

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Kingsolver 1989). Understanding the responses of ectothermic organisms to thermal variability is crucial for predicting the impacts of climate change that can inform policy and conservation decisions (Lee 2023).

Most knowledge regarding responses of ectotherms to thermal change originates from manipulative laboratory experiments conducted at different constant temperatures (Colinet et al. 2015; Massey and Hutchings 2021; Noble, Stenhouse, and Schwanz 2018). Different constant temperature treatments are often implemented to reduce 'noise' and provide an effective approach to compare phenotypic outcomes from exposure to different mean temperatures (Kroeker et al. 2020). However, in natural environments, animals also experience thermal fluctuations around the mean at different timescales (Marshall et al. 2021; Stoks et al. 2017). It is important therefore to determine whether relatively short-term fluctuations around mean temperatures influence phenotypes compared to constant mean temperatures (Bozinovic et al. 2011; Dowd, King, and Denny 2015; Kefford et al. 2022).

Responses of ectotherms to acute temperature change are characterised by non-linear thermal performance curves that can be modelled as a Gaussian function (Huey and Kingsolver 1989). Thermal performance curves vary between species and traits, but common features include an initial increase in performance as temperatures increase from a minimum critical temperature below which function ceases until maximum performance is reached at a thermal optimum. Beyond the thermal optimum there is a rapid decline in performance to a critical thermal maximum beyond which performance ceases (Little and Seebacher 2021; Sinclair et al. 2016). The non-linearity of phenotypic responses to temperature variation means that fluctuating temperatures do not necessarily have the same phenotypic effect as constant temperatures with the same mean (Denny 2019; Marshall et al. 2021; Stoks et al. 2017; Vasseur et al. 2014). These effects of thermal variation can be exacerbated when thermal fluctuations around the mean are irregular (Burton, Lakka, and Einum 2020). Also, different rates of environmental temperature increases can impact phenotypes differently by inducing different levels of oxidative stress (Loughland et al. 2022). Together, these considerations indicate that the shapes of thermal fluctuations can introduce phenotypic differences between fluctuating and constant temperatures with the same mean and that differences increase with increasing amplitudes.

In addition, exposure to different absolute temperature extremes may influence phenotypes. Thermodynamic effects of increasing and decreasing temperatures will influence physiological rate functions acutely, and there may be lasting phenotypic effects of recurring fluctuations if damaging high or low temperatures are reached (Arcus et al. 2016; Knapp and Huang 2022). Hence, it may be predicted that the amplitude of temperature fluctuations impacts phenotypes, with increasing amplitudes causing a decline in performance because of the greater likelihood of reaching damaging temperatures. These effects of increasing amplitudes could be exacerbated by decreasing differences between mean and optimum temperatures, which would increase the likelihood of reaching damaging temperatures. For example, fluctuations lower than optimal temperatures are expected to result in organisms experiencing stressful temperatures less frequently compared to fluctuations that reach temperatures above the optimum (Jørgensen et al. 2022).

Temperature fluctuations may also induce a regulated response by the animal. Temperature fluctuations with long periods (e.g., seasonal fluctuations) often induce thermal acclimation. Thermal acclimation is a regulated process that alters the acute thermal sensitivity of cellular functions and thereby shifts the thermal performance curve, ideally in such a way that potentially negative effects of acute temperature changes are mitigated (Rohr et al. 2018; Seebacher, White, and Franklin 2015). Acclimation and other regulated responses to temperature change (e.g., developmental plasticity) are mediated by a sensory-endocrine-epigenetic axis (Taff et al. 2024; Zimmer, Woods, and Martin 2022). Changes in temperature are transmitted by sensory proteins (e.g., transient receptor potential ion channels) to the central nervous system (e.g., hypothalamus) (Ezquerra-Romano and Ezquerra 2017), which elicits an endocrine response (e.g., mediated by thyroid hormones and glucocorticoids) (Little 2021; Taff et al. 2024) that can change epigenetic programming (e.g., DNA methylation) (Seebacher and Little 2024). However, it is unresolved whether temperature fluctuations with short periods (e.g., diel cycles) can induce regulated responses resembling acclimation. In addition, active transport and conversion between inactive and different active forms of particular hormones modify endocrine signalling and its effects on phenotypes (Little 2021). These dynamics cause variation in phenotypic responses both spatially between tissues and different traits and temporally. For example, glucocorticoids can act rapidly (minutes to hours) to change phenotypes (e.g., metabolic rates) in response to challenging environmental signals (Taff et al. 2024) but are not explicitly known to induce acclimation. Thyroid hormones are well known to regulate acclimation (Little 2021; Little et al. 2013) but the temporal resolution of their action to remodel phenotypes in response to environmental signals is unresolved. The complexity of these mechanisms underlying phenotypic responses to environmental change means that temperature fluctuations may well induce phenotypic changes, but clear directional hypotheses are difficult. However, endocrine signalling varies considerably between taxonomic groups, for example, between invertebrates and vertebrates, and developmental stages (Laudet 2011; Lema 2020). It may be predicted therefore that if the endocrine system is important in mediating phenotypic effects in response to temperature fluctuations there would be phylogenetic differences, as well as differences between different biological responses and life-history stages.

Considering these various mechanisms, it is not surprising that empirical evidence is equivocal (Raynal et al. 2022), leaving the relationship between thermal variability and biological responses unresolved. Our aim therefore was to establish the current state of knowledge regarding the consequences of thermal variability for ectothermic animals. We conducted a meta-analysis to provide a quantitative synthesis of studies that addressed whether biological responses (mean and variance) differed between exposure to constant versus fluctuating thermal environments with the same mean temperature. We tested the hypotheses that (a) the shape of thermal fluctuations around the mean influences biological responses, and irregular fluctuations, in particular, will cause phenotypic outcomes that are different to exposure to a constant mean temperature; (b) on average, larger amplitudes of temperature fluctuations combined with higher mean temperatures will have more pronounced negative effects on phenotypes compared to constant temperature exposure because organisms are more likely to be exposed to stressful temperatures and (c) if endocrine systems are important, biological responses to fluctuating temperatures differ from those to constant temperature exposure because of hormone-mediated effects, leading to differences between traits, life-history stages and taxonomic groups. We first conducted an analysis of the complete data set to determine overall effects and then fitted meta-regression models that included different explanatory groupings (moderators) for which sufficient data were available: (i) biological level of organisation (within-individual traits and population responses), (ii) ecosystems (marine, freshwater and terrestrial), and (iii) exposure periods (acclimation or developmental treatments). Biological responses measured in the literature included within-individual traits such as physiological rates, gene expression and morphology as well as population responses such as survival (full list given below).

2 | Materials and Methods

We followed guidelines set out in the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA; Table S1) (Moher et al. 2015) modified for Ecology and Evolution (PRISMA-Eco Evo) (O'Dea et al. 2021). Question formulation, literature searching and screening were performed according to Foo et al. (2021).

2.1 | Systematic Literature Search and Screening

We followed the PECO (Population, Exposure, Comparator, Outcome) framework (Table S2) (Morgan et al. 2018) to establish the focus questions and assist in the literature search and screening processes. On 5 April 2022, we conducted systematic literature searches in the Scopus, Web of Science and ScienceDirect databases. We searched for studies that investigated the impacts of fluctuating temperatures on ectothermic organisms. Our search included synonyms of 'plasticity', 'acclimation' and 'developmental effects' as we were explicitly looking for treatments imposed within the lifespan of an individual (Noble, Stenhouse, and Schwanz 2018; Pottier et al. 2021). We added 'thermal' and 'temperature' to each search string to limit treatment manipulations to temperature (Barley et al. 2021; Noble, Stenhouse, and Schwanz 2018). Synonyms for 'fluctuating' or 'varying' were then used to specify that one of the treatments must include a thermal variation. The exact search strings used for each database can be found in the Supporting Information *Search strings*.

The search of the three databases returned a total of 13,549 unique studies (Figure S1, Table S2). An initial screening of the abstracts was conducted following the exclusion and inclusion criteria (Figure S2). Briefly, the study had to be in English and experimentally compare the effects of a constant and a fluctuating temperature treatment on an ectothermic organism in controlled laboratory conditions. We limited studies to controlled laboratory conditions because the treatment effects we were interested in (stable vs. fluctuating temperatures) would be nearly impossible to achieve under field conditions. We did not place any a priori limits on the duration of fluctuations. However, to ensure that there were differences between control and treatment conditions, fluctuations must have had a minimum total range of 1.5°C (amplitude = 0.75°C) and the difference between means of constant and fluctuating treatments must have been less than 1°C (Table S2). There were 202 studies that matched these criteria. We conducted a full-text screening of these studies (Figure S2). For inclusion in the analysis, each study had to have (a) treatments with identical housing and habituation periods, (b) fluctuating temperatures imposed during the lifespan of an individual (acclimation and developmental effects), (c) constant and fluctuating treatments with the same mean temperature and all other conditions identical and (d) report a biological response, excluding metrics of thermal performance such as critical thermal minima or maxima. Studies had to have all relevant statistical information (including sample size, mean and error estimates). Principal components and factor loadings were excluded (Tarka et al. 2018). In total, there were 57 studies that adhered to both screening protocols and were included in the analysis. All authors (except DWAN) conducted screening of the studies and completed both stages using Rayyan software (Ouzzani et al. 2016).

Next, from the 57 studies selected above papers citing the original study (forward search), and papers that were cited in the original study (backward search), were identified in Scopus, Web of Science, ScienceDirect and Google Scholar on 19 October 2022. This process yielded a total of 3932 unique studies that also underwent abstract and full-text screenings. As a result, an additional 87 studies were included. There was one species in the compiled dataset with an unresolved phylogeny in the Open Tree of Life database, which required the exclusion of the study (Michonneau, Brown, and Winter 2016; Slein et al. 2023). At the conclusion of the systematic literature search and screening process, there were a total of 143 studies which were included in the analysis (Supporting Information *List of included studies*).

2.2 | Data Extraction

From papers in the final selection, we extracted data (sample size, mean and error metric) manually from text, tables, Supporting Information and figures (using R package *metaDigitise* to extract data from Figures) (Pick, Nakagawa, and Noble 2019). Biological replicates were used to determine sample size, rather than technical replicates, to avoid non-independence (Wu and Seebacher 2020). Constant temperature treatments were considered the 'control' that was compared to a fluctuating treatment. When there were multiple fluctuating treatments, each treatment was compared to its appropriate control (Raynal et al. 2022). For studies that conducted paired control and fluctuating treatments across multiple mean temperatures, all comparisons were recorded (Raynal et al. 2022). If there were multiple acute test temperatures at which biological responses were measured, we selected those that coincided with the treatment mean (Seebacher, White, and Franklin 2015). In addition, we recorded species information (taxonomy, habitat and source), experimental design details (exposure type, length and life-history stage), details of the fluctuation treatment (mean temperature, amplitude, number of fluctuations, fluctuation type and length), and phenotypic outcome (response category, measurement and units). All authors (except DWAN) were involved in the data extraction process, and we ensured consistency between authors by extracting data collectively from a subset of papers. To remove any conflict of interest, authors were not permitted to screen or extract studies they had a previous association with (Macartney, Lagisz, and Nakagawa 2022).

2.3 | Data Processing and Transformations

Standard errors (SE) and 95% confidence intervals (CI) were converted to standard deviations (SD) (Quinn and Keough 2002) for effect size calculations. Data reported as percentages (e.g., survival) were arcsine-transformed according to Macartney, Lagisz, and Nakagawa (2022), and means and variances that were reported on the natural log scale were transformed according to Higgins, White, and Anzures-Cabrera (2008). To avoid taking the natural log of zero during effect size calculations, a constant of 0.5 was added to all means and SD (Rubin and Seebacher 2022; Wu and Seebacher 2020).

In cases where different terminology was used in different publications, we categorised life history stages and traits for consistency in the analysis (Tables S3–S18). Species names were adjusted to match the Open Tree of Life database (using the function *tnrs_match_names* to consolidate and correct scientific names; Michonneau, Brown, and Winter 2016) from which taxonomic information was retrieved. Coordinates of sampling sites for wild-sourced organisms were categorised into tropics, temperate and polar regions (Lønborg et al. 2021). Amphibians were analysed as aquatic, freshwater organisms.

2.4 | Effect Size Calculations

We calculated the log response ratio (Equation 1) as the effect size (Hedges, Gurevitch, and Curtis 1999; Lajeunesse 2011) and the sampling variance (Equation 2) as in (Nakagawa et al. 2017):

$$\ln\text{RR} = \ln\left(\frac{\bar{x}_F}{\bar{x}_C}\right) \tag{1}$$

$$v_{\ln\text{RR}} = \frac{\text{SD}_F^2}{n_F\bar{x}_F^2} + \frac{\text{SD}_C^2}{n_C\bar{x}_C^2} \tag{2}$$

where n is the sample size, $\bar{x}$ is mean, and subscripts C and F denote the constant and fluctuating temperature treatments, respectively. For studies that had a low sample size ($n=2$), we applied the Lajeunesse Delta method (Equation 3 for effect size and Equation 4 for the sampling variance; Lajeunesse 2015):

$$\ln\text{RR}^{\Delta} = \ln\text{RR} + \frac{1}{2}\left(\frac{\text{SD}_F^2}{n_F\bar{x}_F^2} - \frac{\text{SD}_C^2}{n_C\bar{x}_C^2}\right) \tag{3}$$

$$v_{\ln\text{RR}^{\Delta}} = v + \frac{1}{2}\left(\frac{\text{SD}_F^4}{n_F^2\bar{x}_F^4} + \frac{\text{SD}_C^4}{n_C^2\bar{x}_C^4}\right) \tag{4}$$

where notation is the same as Equations (1 and 2). lnRR was used rather than the standardised mean differences (SMD), as it is more robust to heteroscedasticity between treatment groups, often provides greater statistical power, and allows for an easier comparison with lnCVR (Macartney, Lagisz, and Nakagawa 2022; Nakagawa et al. 2015; Yang et al. 2022). Effect sizes were calculated so that a negative value represents a detrimental impact of fluctuating temperatures on the organism (Tables S12–S18) (Hasik et al. 2023; Slein et al. 2023).

To investigate differences in phenotypic variation between the two treatments, we calculated and analysed the log coefficient of variation ratio (lnCVR; Equation 5 shows effect size and Equation 6 the sampling variance; Senior, Viechtbauer, and Nakagawa 2020):

$$\ln\text{CVR} = \ln\left(\frac{\text{CV}_F}{\text{CV}_C}\right) + \frac{1}{2}\left(\frac{1}{n_F-1} - \frac{1}{n_C-1}\right) + \frac{1}{2}\left(\frac{\text{SD}_C^2}{n_C\bar{x}_C^2} - \frac{\text{SD}_F^2}{n_F\bar{x}_F^2}\right) \tag{5}$$

$$v_{\ln\text{CVR}} = \frac{\text{SD}_C^2}{n_C\bar{x}_C^2} + \frac{\text{SD}_C^4}{2n_C^2\bar{x}_C^4} + \frac{n_C}{(n_C-1)^2} + \frac{\text{SD}_F^2}{n_F\bar{x}_F^2} + \frac{\text{SD}_F^4}{2n_F^2\bar{x}_F^4} + \frac{n_F}{(n_F-1)^2} \tag{6}$$

where notation is the same as Equations (1 and 2).

2.5 | Statistical Analysis

All calculations, analyses and figure production were completed with R.4.4.0 in RStudio version 2024.04.1 (using R Package *ggplot2* for figures; Wickham 2011). All data and code are available at https://github.com/ClaytonStocker/Fluctuation_Meta (Table S19). Versions of all software packages used are given in Table S169. Data are presented as both the mean effect sizes ±95% CIs and as a percentage difference between treatments (exponentially transformed mean lnRR or lnCVR; Pustejovsky 2018).

In the analyses, we included the following random effects to account for dependencies in our dataset (Nakagawa et al. 2017; Sánchez-Tójar et al. 2020): Study ID, to account for several effect sizes coming from the same experiment; and Shared Animal, to account for multiple responses measured in the same individuals. To control for phylogenetic relatedness, a phylogenetic tree (Figure S3) was created using the Open Tree of Life database (Michonneau, Brown, and Winter 2016), where polytomies were resolved randomly (using R Package *Ape*; Paradis, Claude, and Strimmer 2004) and Grafen's method was used to calculate branch lengths (Power = 1; Grafen 1989). For inclusion in the model, the phylogenetic tree was converted into a correlation/covariance matrix. Additionally, the repeated use of a species across studies was accounted for with a species-level random effect. We also added a trait-level random effect to reduce the chance of certain traits dominating the analysis (excluding meta-regressions with specific biological responses as the moderator). To estimate the final 'residual' variation, we added an observation-level random effect (Nakagawa and Santos 2012).

Frequentist models were employed, using the restricted maximum likelihood (REML) approach with a t-statistic containing adjusted degrees of freedom based on the number of studies, and an adjusted convergence threshold ($1e^{-8}$; Nakagawa et al. 2021). Models were fitted with a modified sampling covariance matrix to account for instances where several fluctuating treatments were compared to the same constant temperature treatment (Noble et al. 2022).

2.6 | Multi-Level Meta-Analytic Models

Multi-level meta-analytic (MLMA) models were fitted using the *metafor* package in R (using the function *rma.mv*; Viechtbauer 2010). Firstly, an MLMA model was implemented to estimate the overall meta-analytic mean of the complete dataset. Subsequently, we conducted several subset analyses to estimate the meta-analytic means for population responses and within-individual traits, wild organisms sampled from temperate latitudes (tropics and polar regions had too few effect sizes for analysis), terrestrial and aquatic organisms (marine and freshwater) and acclimation or developmental exposure (Figure 1). The definitions of included terms in each subset can be found in Table 1. From our MLMA models, the extended heterogeneity statistic (I^2) was calculated to quantify unexplained variation after removing the known sampling variance. The total heterogeneity (I^2_{Total}) was partitioned to correspond with each random effect (I^2_{Animal}, $I^2_{\text{Measurement}}$, I^2_{Obs}, I^2_{Phylo}, I^2_{Study} and I^2_{Species}) (Tables S20 and S21) (Borenstein et al. 2021; Nakagawa and Santos 2012). I^2 of 75% is considered high, 50% medium and 25% small (Higgins et al. 2003). However, in multispecies meta-analysis, it is not uncommon to achieve an I^2 that is greater than 90% (Senior et al. 2016).

2.7 | Meta-Regression Models

Meta-regression models were conducted using a variety of moderators that had the potential to influence the observed mean and variance effect size estimates. Categorical moderators included developmental exposure period (embryo, larva, pupa and juvenile), exposure type (acclimation and developmental), fluctuation type (sinusoidal, alternating, stepwise and stochastic; Figure 2), acclimation life-history stages (embryo, larva, pupa, juvenile and adult), biological response categories (behavioural, biochemical assay, gene expression, life history, morphological and physiological), specific biological responses (see Tables S12–S18 for a full list of biological responses), and taxonomic class (see Figure S3 for phylogenetic tree). Levels within categorical moderators were only included if there were >10 effect sizes (O'Dea et al. 2021). Most of the studies included used diel cycles as their fluctuating treatment, and for comparisons between diel cycles and fluctuations with a different period, we grouped non-diel cycles into 'shorter' cycles of ≤8 h and 'extended' cycles of ≥48 h. Additionally, we investigated continuous moderators (Figure 1), including the amplitude of fluctuations (expressed as a total thermal range [i.e., 2×amplitude] = 1–45.5°C), and the number of fluctuations (treatment exposure duration/period of fluctuations; range = 4–546 fluctuations). We tested for a significant interaction between the amplitude of fluctuations and mean treatment temperatures to determine whether higher mean temperatures and amplitudes acted synergistically to decrease trait values.

2.8 | Publication Bias and Sensitivity Analysis

To investigate significant publication bias in our results, we firstly conducted a visual inspection of funnel plots for indications of asymmetry (Sterne, Becker, and Egger 2005). The presence of a time-lag effect (or decline effect) was assessed by adding z-transformed publication year and precision (inverse of variance) as moderators to our overall MLMA model (Nakagawa et al. 2022). A time-lag bias would be evident if 95% confidence intervals of the slope were to exclude zero (Wu and Seebacher 2020). Briefly, the investigation into publication bias found evidence of time-lag bias in our lnRR data set, with a significant increase in mean biological responses in more recent years. There was no evidence of publication bias in our lnCVR data set (Supporting Information *Publication bias and sensitivity analysis*).

A sensitivity analysis to test the 'robustness' of our results by estimating the influence of individual data points on our overall MLMA model (using the function *cooks. distance* in the R Package *metafor*; Viechtbauer 2010). The results of the overall MLMA model were also compared to models fitted with a comparable effect size (Hedge's *g*; Gurevitch, Morrison, and Hedges 2000) and untransformed data (Knol et al. 2011; Macartney, Lagisz, and Nakagawa 2022). The findings from the sensitivity analysis suggested that both the lnRR and lnCVR data sets were robust and free from significantly influential outliers (Supporting Information *Publication bias and sensitivity analysis*).

In addition to these analyses, robust variance estimators (RVE) (using the function *robust* in the R package *metafor*; Viechtbauer 2010) were used to provide insights into the sensitivity of results to complex forms of non-independence that can inflate type I error rates. While mean estimates are not impacted by RVE, simulations have shown that they can correct inflated error rates at the nominal level (Nakagawa et al. 2021). Mean effect size estimates for gene expression measurements (except for temperate organisms), the effect of thermal range on marine organisms, the interaction between thermal range and mean temperature in population responses, and differences in the variance of biological responses for wild organisms from temperate latitudes remained significant following robust variance estimates. Below, we report results from standard variance estimates, but in our interpretation, we focus on estimates that remained significant after applying RVE.

3 | Results

3.1 | Do Mean Biological Responses Differ Between Constant and Fluctuating Environments?

The final overall dataset included 1492 effect sizes from 143 studies across 118 species. 89.7% of effect sizes (130 studies, 1333 effect sizes) stemmed from studies that used diel fluctuations as the experimental (fluctuating) treatment, 1.9%

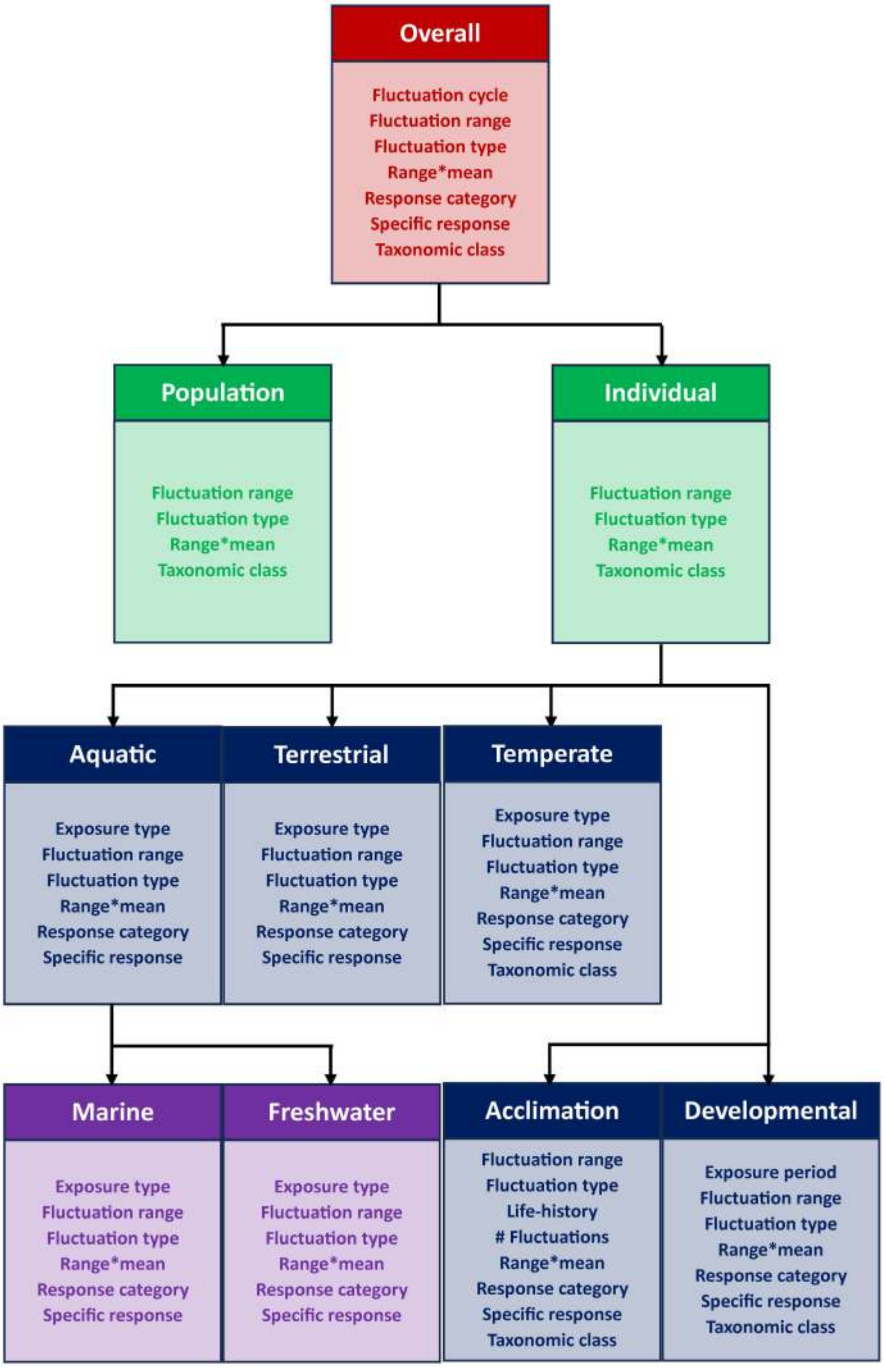

FIGURE 1 | Analysis strategy. The order of subset MLMA analyses and the corresponding meta-regression analyses (white text = MLMA model; coloured text = meta-regression model). 'Fluctuation cycle' distinguishes between shortened, diel and extended cycles. 'Fluctuation range' refers to the amplitude expressed as the total thermal range (i.e., 2 × amplitude) experienced during the fluctuation treatment. 'Fluctuation type' denotes the thermal regime implemented in the experimental design (Figure 2). 'Range and Mean' is the interaction between fluctuation range and thermal mean. 'Specific response' and 'Response categories' refer to the within-individual traits and categorised groups of traits measured following treatment exposure. 'Taxonomic class' is the class of each species. 'Exposure type' distinguishes between fluctuations imposed during developmental periods or acclimation treatments. Acclimation 'Life history' and Development 'Exposure periods' refer to the life-history stage during experimental treatments. 'Number of fluctuations' is the treatment exposure duration divided by the period of each fluctuation. Definitions of MLMA and meta-regression terms are found in Table 1.

of effects sizes stemmed from studies using shorter cycles (4 studies, 28 effect sizes) and 8.4% (11 studies, 124 effect sizes) from extended cycles. Overall, the means of the biological responses in constant and fluctuating thermal environments were not significantly different (Figure 3), and there was no difference between diel cycles, shorter cycles, and extended cycles in their effects on phenotypes (moderator = fluctuation cycle; Figure S18).

TABLE 1 | Definitions of terms in each subset. An outline of the requirements for the inclusion of effect sizes into each subset analysis.

Subset analyses	Definitions
Population responses	Biological responses measured for an entire population (e.g., survival)
Within-individual traits	Biological responses measured in an individual (e.g., Food Consumption, Cortisol Levels, hsp70, Development Time, Length, Locomotor Performance, etc.)
Temperate organisms	Wild organisms that were collected from temperate latitudes
Aquatic organisms	Organisms that spend the predominant amount of their lifecycle in aquatic ecosystems, including amphibians
Marine organisms	Organisms that spend the predominant amount of their lifecycle in marine ecosystems
Freshwater organisms	Organisms that spend the predominant amount of their lifecycle in freshwater ecosystems
Terrestrial organisms	Organisms that spend the predominant amount of their lifecycle in terrestrial ecosystems
Acclimation	A treatment that does not encompass a significant period of development, and a reversible biological response is measured. This generally appears in an experimental design with a treatment and trait measurement within the same life-history stage
Developmental	A treatment exposure that encompasses a significant developmental period. This generally appears in an experimental design with treatments and trait measurements across life-history stages

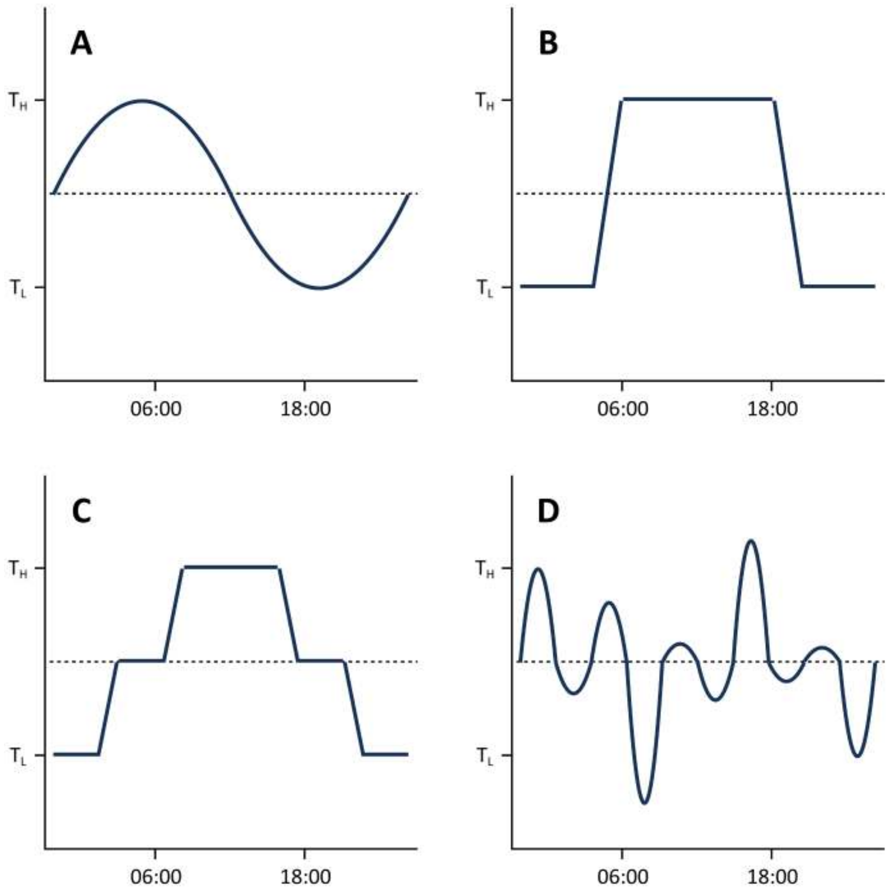

FIGURE 2 | Examples of different fluctuation types in the experimental designs of included studies. Fluctuation patterns included sinusoidal regimes (A), alternating regimes (B), stepwise regimes (C) and stochastic regimes (D). T_H and T_L are high and low experimental temperatures, respectively. Time units presented are examples of fluctuating regimes imposed with a diel cycle (89% of effect sizes).

Total heterogeneity was high ($I^2_{Total} = 99.69\%$), with little to no variation explained by the biological responses ($I^2_{Measurement} = 1.57\%$), phylogeny ($I^2_{Phylo} = 0.00\%$) and species ($I^2_{Species} = 0.00\%$). Together, shared animal ($I^2_{Animal} = 25.70\%$), observational ($I^2_{Obs} = 56.48\%$) and study ($I^2_{Study} = 15.94\%$) random effects explained most of the heterogeneity observed (heterogeneity statistics for each subset analyses in Table S20).

Meta-regression analyses showed that measurements of gene expression were significantly different between the two thermal

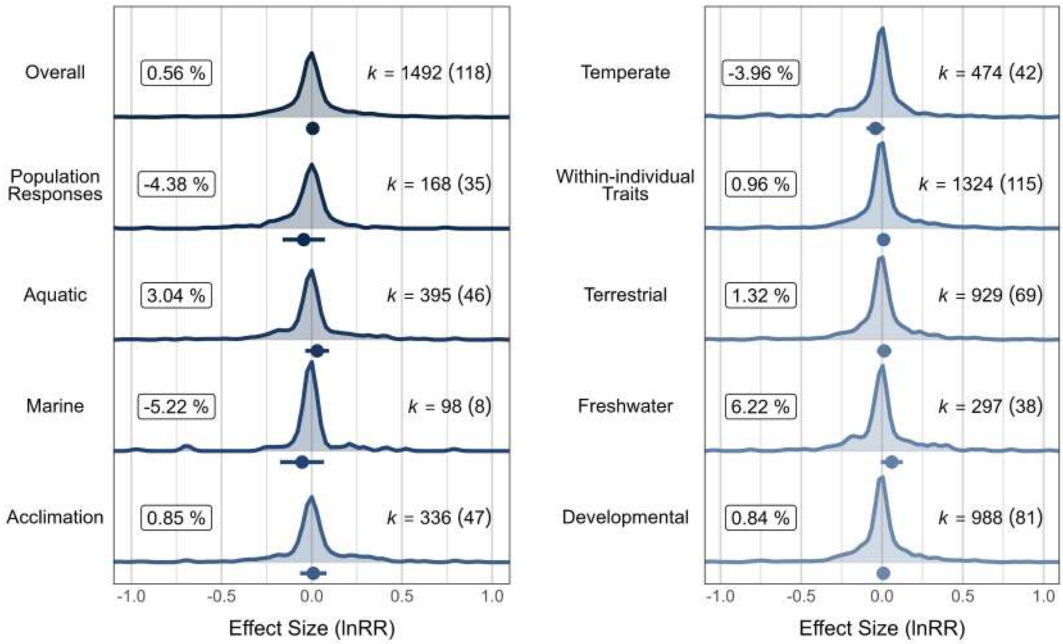

FIGURE 3 | MLMA results for the overall and subset datasets. There were no significant differences in mean trait values between the thermal treatments (effect size 95% CIs crossing 0) in the overall model or the subsets. Mean effect size (lnRR) estimates ±95% CIs (solid circles and horizontal bars) are shown, as well as distributions of individual effect sizes. Percentage labels are the mean lnRR estimates transformed to show a proportional difference between fluctuating and stable temperature treatments. *k* = number of effect sizes with the number of species in brackets. *X*-axis limits are cropped for presentation.

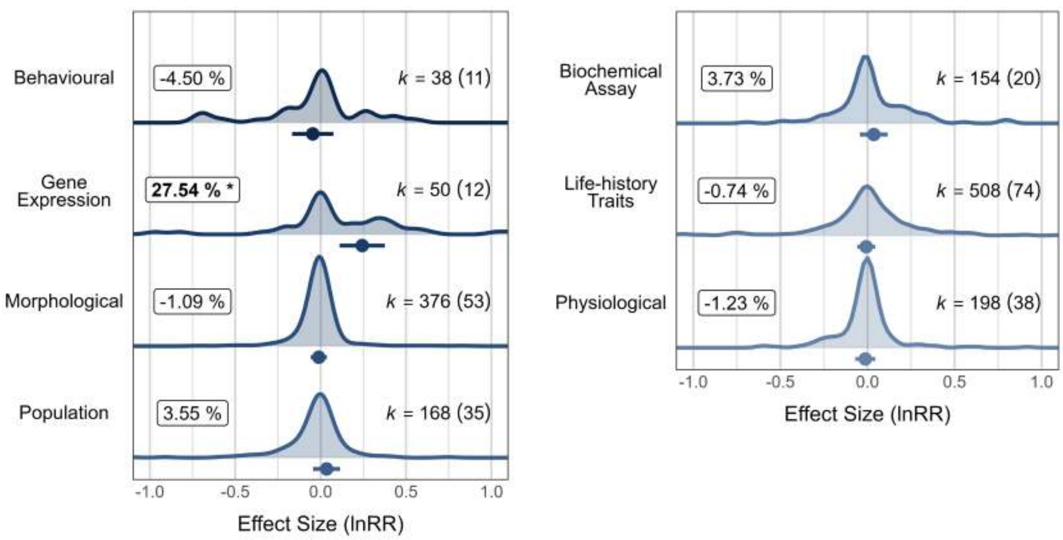

FIGURE 4 | Meta-regression results for the overall data set with biological response category as the moderator. There was a significant increase in gene expression found in fluctuating treatments, indicated in the figure by an asterisk next to the bold percentage label ($p < 0.05$). Mean effect size (lnRR) estimates ±95% CIs (solid circles and horizontal bars) are shown, as well as distributions of individual effect sizes. Percentage labels are the mean lnRR estimates transformed to show a proportional difference between the fluctuating and constant temperature treatments. *k* = number of effect sizes with the number of species in brackets. *X*-axis limits are cropped for presentation.

treatments (moderator = biological response category; effect size estimate = 0.24; 95% CIs = [0.11, 0.38]; $p < 0.001$; $n = 50$; Figure 4), driven mostly by the expression of the hsp70 gene (moderator = specific biological responses; effect size estimate = 0.40; 95% CIs = [0.14, 0.66]; $p < 0.001$; $n = 12$; Figure 5). The increase in gene expression in fluctuating treatments was significant in temperate (38.8% increase, 42 species total), aquatic (38.1% increase, 46 species) and freshwater environments (39.2% increase, 38 species total) and during developmental treatments (23.4% increase, 8 species; Table 2).

The longevity of an organism was significantly less in the fluctuating compared to constant temperature treatments (moderator = specific biological responses; effect size estimate = −0.10; 95% CIs = [−0.17, −0.02]; $p = 0.01$; $n = 93$; Figure 5). Meta-regression analyses testing for the effect of taxonomic class

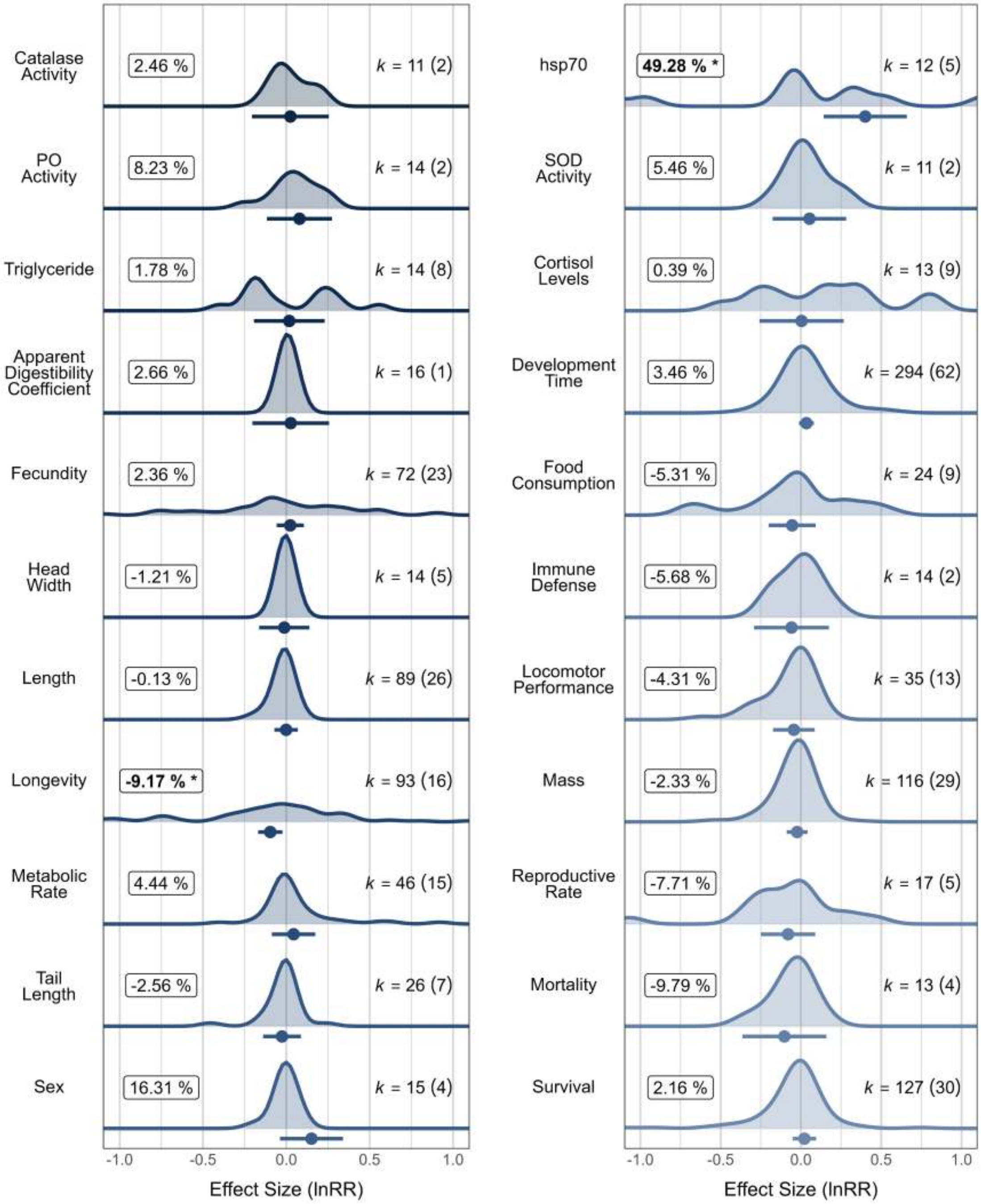

FIGURE 5 | Meta-regression results for the overall data set with specific biological responses as the moderator. There was a significant increase in the expression of the hsp70 mRNA and a decrease in longevity found in the fluctuating temperature treatment, indicated in the figure by an asterisk next to the bold percentage label ($p < 0.05$). Mean effect size (lnRR) estimates ±95% CIs (solid circles and horizontal bars) are shown, as well as distributions of individual effect sizes. Specific biological responses are ordered thematically (molecular, cellular, whole animal, and population responses). Percentage labels are mean lnRR estimates transformed to show a proportional difference between the fluctuating and constant temperature treatments. k = number of effect sizes with the number of species in brackets. X-axis limits are cropped for presentation.

(Figure 6) and fluctuation type (Figure 7) as moderators showed no significant differences between constant and fluctuating treatments.

The thermal range (2 × amplitude) of the fluctuations had a significant effect on the mean biological responses overall (moderator = fluctuation range, effect size estimate = −0.01; 95% CIs = [−0.01, 0.00]; $p = 0.01$; $n = 1492$). As the thermal range of the fluctuations increased, the mean biological response values decreased relative to the stable treatment (Figure 8), particularly in population responses, temperate organisms, and in aquatic organisms (particularly in marine environments; Table 2). There were significant interactions between thermal range and mean treatment temperatures influencing responses to fluctuating environments in the following categories: population responses, within-individual traits, terrestrial

TABLE 2 | Results of the subset meta-regression analyses that found significant differences in mean biological response values between the constant and fluctuating thermal treatments. Effect size = lnRR.

Subset	Moderator	Estimate	CI low	CI high	p
Population	Fluctuation range	−0.03	−0.04	−0.02	0.00
	Range × mean	0.01	0.00	0.01	0.00
Within-individual	Range × mean	0.00	0.00	0.00	0.00
Temperate	Fluctuation range	−0.01	−0.01	0.00	0.00
	Gene expression	0.33	0.14	0.52	0.00
	Life-history traits	−0.18	−0.29	−0.07	0.00
Aquatic	Alternating regime	0.10	0.02	0.18	0.02
	Fluctuation range	−0.00	−0.01	−0.00	0.04
	Food consumption	−0.16	−0.30	−0.01	0.03
	Gene expression	0.32	0.20	0.45	0.00
	Mass	−0.09	−0.14	−0.04	0.00
Marine	Fluctuation range	−0.01	−0.02	0.00	0.04
Freshwater	Alternating regime	0.10	0.00	0.20	0.05
	Gene expression	0.33	0.20	0.46	0.00
	Mass	−0.09	−0.14	−0.04	0.00
Terrestrial	Range × mean	0.00	0.00	0.00	0.00
Developmental	Gene expression	0.21	0.05	0.37	0.01
	Range × mean	0.00	0.00	0.00	0.00

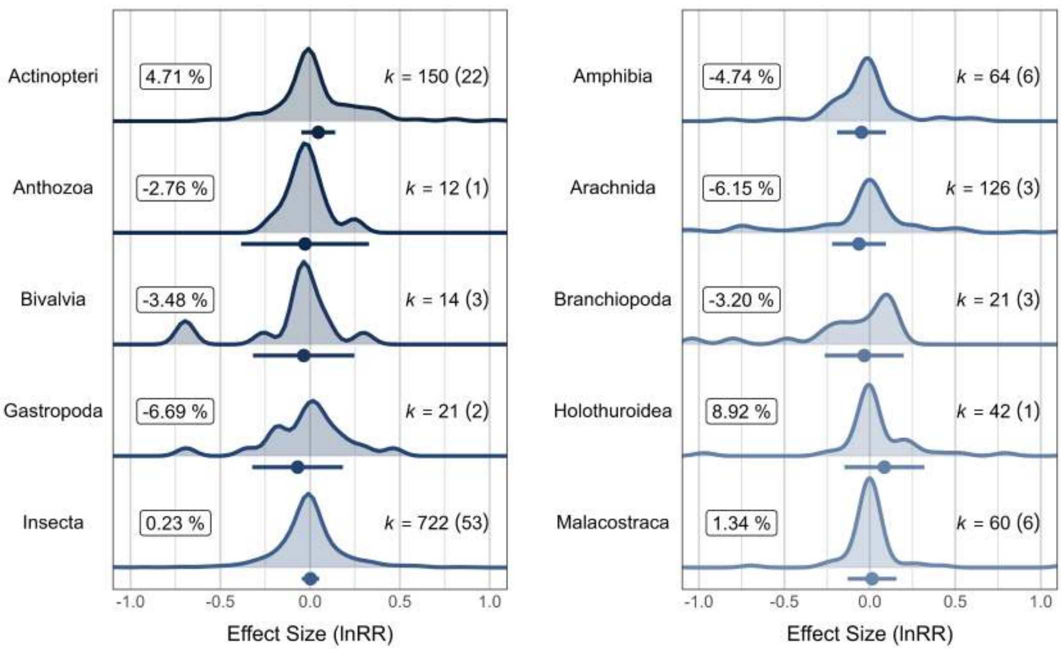

FIGURE 6 | Meta-regression results for the overall data set with the taxonomic class as the moderator. There were no significant differences in mean trait values between the thermal treatments (effect size 95% CIs crossing 0). Mean effect size (lnRR) estimates ±95% CIs (solid circles and horizontal bars) are shown, as well as distributions of individual effect sizes. Classes ordered taxonomically (vertebrates and invertebrates). Percentage labels are mean lnRR estimates transformed to show a proportional difference between fluctuating and stable temperature treatments. k = number of effect sizes with the number of species in brackets. X-axis limits are cropped for presentation.

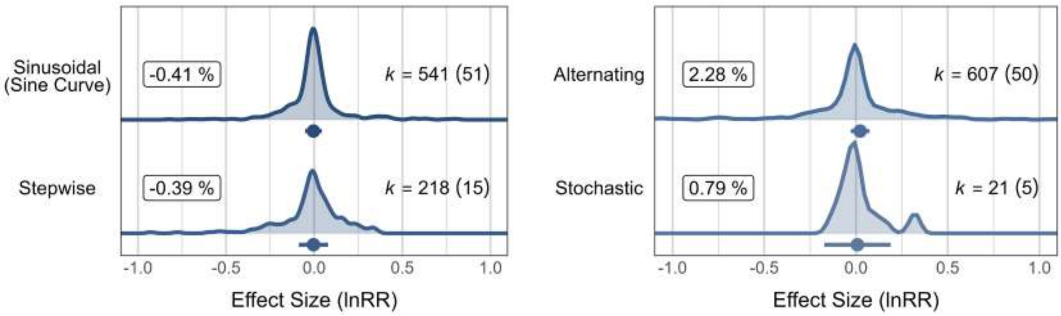

FIGURE 7 | Meta-regression results for the overall data set with fluctuation type as the moderator. There were no significant differences in mean trait values between the thermal treatments (effect size 95% CIs crossing 0). Mean effect size (lnRR) estimates ±95% CIs (solid circles and horizontal bars) are shown, as well as distributions of individual effect sizes. Percentage labels are mean lnRR estimates transformed to show a proportional difference between fluctuating and stable temperature treatments. k = number of effect sizes with the number of species in brackets. X-axis limits are cropped for presentation.

organisms, developmental treatments (moderator = interaction of fluctuation range and thermal mean; Table 2).

Subset analyses of aquatic organisms (10.5%, 16 species), particularly in freshwater environments (10.4%, 15 species), indicated a significant increase in mean biological response values under the fluctuation type "alternating thermal regime" compared to stable temperatures (moderator = fluctuation type; Table 2). Additionally, in aquatic organisms, there was a decrease in food consumption (−14.4%, 6 species) and mass (−8.6%, 7 species, particularly in freshwater environments, −8.6%, 7 species) in fluctuating compared to constant temperature treatments (moderator = specific biological responses; Table 2). Life-history trait measurements were significantly different between the two thermal treatments (−16.5%, 22 species) in wild organisms sampled from temperate latitudes (moderator = biological response category; Table 2). All other subset meta-regression analyses showed no significant differences in mean trait values between thermal treatments.

Full results of all analyses are shown in Tables S26–S96.

3.2 | Do Variances of Biological Responses Differ Between Constant and Fluctuating Temperatures?

Both the overall and subset MLMA models indicated no significant differences in biological response variance between constant and fluctuating thermal treatments (Figure S11). There was a medium level of total heterogeneity ($I^2_{\text{Total}} = 69.32\%$), with little to no variation being explained by the shared animal ($I^2_{\text{Animal}} = 0.00\%$), biological response ($I^2_{\text{Measurement}} = 0.00\%$), phylogeny ($I^2_{\text{Phylo}} = 0.00\%$), species ($I^2_{\text{Species}} = 0.00\%$), and study ($I^2_{\text{Study}} = 4.86\%$) random effects. Most heterogeneity was explained by 'residual' variation ($I^2_{\text{Obs}} = 64.45\%$; heterogeneity statistics for each subset analysis are shown in Table S21).

The variance of biological responses increased under the fluctuation type "alternating thermal regime" (moderator = fluctuation type, effect size estimate = 0.07; 95% CIs = [0.02, 0.12]; $p = 0.01$; $n = 237$; Figure S11), and in Arachnida (moderator = taxonomic class, effect size estimate = 0.12; 95% CIs = [0.05, 0.18]; $p = 0.00$; $n = 75$; Figure S12). An increase in the range of temperature fluctuations was significantly associated with increases in the variance of biological response values for temperate organisms (moderator = fluctuation range, effect size estimate = 0.00; 95% CIs = [0.00, 0.001]; $p = 0.00$; $n = 474$; Figure S13). Similar to lnRR above, there were significant interactions between thermal range and mean treatment temperatures influencing responses to fluctuating environments in population responses, within-individual traits, terrestrial organisms, developmental treatments, and in the analysis of the overall model (moderator = interaction of fluctuation range and thermal mean; Table S97).

Meta-regression analyses indicated that the variance in life-history traits of aquatic organisms differed significantly between the two thermal treatments (moderator = biological response category, effect size estimate = −0.16; 95% CIs = [−0.25, −0.07]; $p < 0.001$; $n = 67$; Figure S14), mostly driven by measurements of development time (moderator = specific biological response, estimate = −0.17; 95% CIs = [−0.28, −0.06]; $p < 0.001$; $n = 35$; Figure S15). Life-history traits (−11.7%, 17 species; Figure S16) and development time (−15.3%, 15 species; Figure S17) were less variable in fluctuating compared to constant temperatures in freshwater environments.

All other meta-regression analyses of the overall and subset datasets showed no significant effect of fluctuating temperatures on biological response variance (Tables S98–S168).

4 | Discussion

Our meta-analysis of the entire dataset shows that on average fluctuating temperatures do not alter the mean or variance of biological responses compared to constant temperatures. The dataset included measurements across a range of within-individual traits and population responses, ecosystems, and types of exposure. Subset analyses of these more specific contexts also showed limited effects of temperature fluctuations on mean and variance of responses. However, there are special cases where temperature fluctuations did make a difference and we discuss these below. We also note that the studies we considered reported laboratory experiments with controlled constant and variable conditions. This inclusion criterion was necessary so that the variable treatment itself and its effect

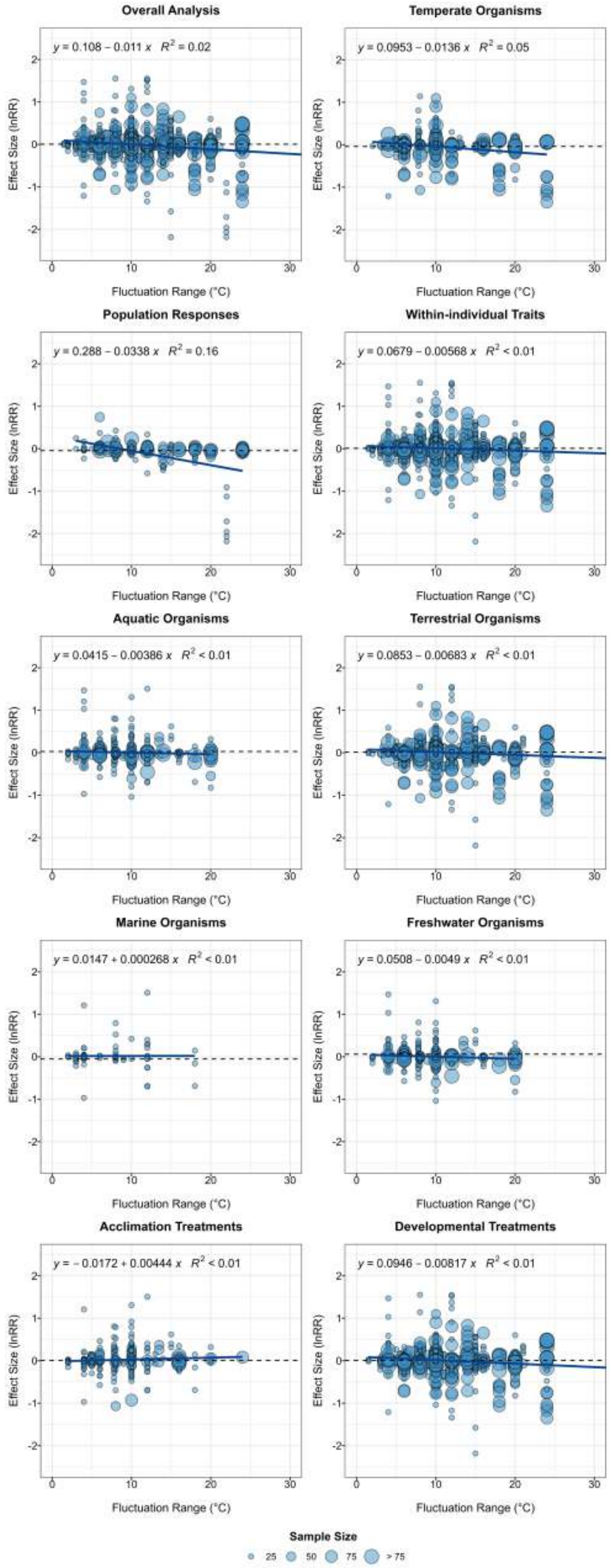

FIGURE 8 | Relationship between effect size (lnRR) and the range of the temperature fluctuations. The overall and subset analyses for temperate organisms, population responses and aquatic (specifically marine) organisms showed that an increase in the range of the temperature fluctuations was significantly associated with decreases in the mean biological response values. Dashed line = lnRR estimate from the overall MLMA model. Solid line = model prediction. Sample sizes are those used to calculate each individual effect size. Axes were cropped for presentation.

could be clearly quantified. It is almost impossible to find or control field conditions so that there are clearly delineated constant and variable treatments. Laboratory studies are important because they show in-principle responses, but we note that these responses could be modified by more complex environmental interactions in the field (see also below). Additionally, most studies in our analysis used diel fluctuations. Our analysis shows that fluctuation period did not influence responses, but the relatively small sample sizes for shorter and extended cycles would warrant further investigation into the potential effects of fluctuation periods (Breitenbach et al. 2020).

Climate change is predicted to manifest as an increase in temperature means, changing fluctuation amplitudes, and in the frequency of extreme climatic events (Pachauri et al. 2014; Vasseur et al. 2014). Short-term, erratic thermal variations such as heat waves can impact the phenotypes of ectotherms (Bauerfeind and Fischer 2014; Breitenbach et al. 2020; Johansen et al. 2021). Heat waves can create temporary thermal conditions that can have long-term detrimental consequences. For example, concurrent exposure to elevated temperature and pathogen load caused increased mortality in a frog (*Atelopus zeteki*), and the combination of these stressors is likely to have caused declines in the species' natural range (Cohen et al. 2019). However, findings from our meta-analysis indicate that recurring fluctuations around a constant mean have a limited effect on biological responses, which are instead more likely to be determined by mean temperatures.

Nonetheless, the magnitude of fluctuation matters for some traits, and increasing amplitudes of fluctuations decreased mean trait values relative to the constant thermal treatment. Similarly, previous analyses showed that various forms of acclimation, and development in reptiles were affected negatively by increasing temperature when variability was high (Raynal et al. 2022; Slein et al. 2023). Even if the immediate effects of increasing amplitudes are relatively small, the long-term ecological effect could be relevant by, for example, decreasing survival. Population responses were influenced by the interaction between mean temperatures and temperature fluctuation amplitudes. Interestingly, however, increasing mean temperatures had a positive effect on trait values while increasing fluctuations reduced trait values. Temperature mean and amplitude therefore did not act synergistically. Our results indicate that mean values of temperature treatments in the literature were on the left-hand side of the optimal temperature in thermal performance curves so that an increase in temperature also caused an increase in performance (up to the optimal temperature). However, increasing amplitudes are likely to have pushed temperatures to above-optimal temperatures. Increases beyond optimal temperatures can cause cellular damage, and thereby a decrease in performance.

Extreme fluctuations could expose organisms to body temperatures near the critical thermal limits of their performance curves and therefore have a detrimental impact on performance (Amarasekare and Savage 2012; Jørgensen et al. 2022; Schulte, Healy, and Fangue 2011). We found that mRNA levels ('gene expression') increased in fluctuating temperatures (overall data set, aquatic organisms, and developmental exposure), which was mostly driven by the expression of hsp70. Increased hsp70 is indicative of a stress response, and functions in facilitating protein re-folding and preventing protein degradation in stressful environments (Shamovsky and Nudler 2008; Vasseur et al. 2014), while the continued over-expression or dysregulation of hsp70 can result in pathological conditions (Peraçoli et al. 2013; Wu et al. 2019). Ectotherms are particularly susceptible to thermal variability and may experience thermal stress as a result of temperature fluctuations, particularly at higher amplitudes (Chown et al. 2010; Jørgensen et al. 2022), and increased hsp70 expression may help stabilise cellular biochemistry. Increased hsp70 in fluctuating environments can indicate thermal stress, which may also be reflected in the decrease in longevity (overall data set), food consumption (aquatic organisms) and body mass (aquatic organisms) we observed. Stressful temperature fluctuations may influence the optimal allocation of resources that can impact life history and physiological processes (McNamara and Buchanan 2005; Monaghan, Metcalfe, and Torres 2009). Trade-offs in resource allocation could compromise growth and ageing (Monaghan, Metcalfe, and Torres 2009; Zera and Harshman 2001), and these relationships are an important direction for future research.

Manipulative experiments conducted at constant laboratory temperatures are the foundation for understanding how ectothermic organisms respond to thermal change (Colinet et al. 2015; Massey and Hutchings 2021; Noble, Stenhouse, and Schwanz 2018). The state of knowledge we established indicates that longer-term (i.e., over 4+ diel cycles) biological responses of ectotherms are primarily driven by mean environmental temperatures. We note, however, that most studies in our analysis implemented diel cycles and that critical thermal limits (which we did not consider here) may be more strongly influenced by temperature fluctuations (Jørgensen et al. 2021). Regulating phenotypes in response to regular, short thermal fluctuations such as daily variation could be disadvantageous for ectotherms by producing excessive phenotypic variation. Instead, a time lag could reflect a beneficial response that reduces phenotypic variation. A possible time lag would most likely be mediated by neuro-endocrine signalling as we discussed in the Introduction. Our analysis indicates that diel fluctuations, which most studies implemented in their experiments, are not sufficient to induce regulated changes in phenotype or acclimation. The characteristics of the environmental signals that induce acclimation are as yet unresolved and are likely to differ between different taxa and traits. It is also possible that body size may influence the rate of onset of acclimation responses (Rohr et al. 2018). Nonetheless, stressful temperature exposures can elicit relatively rapid phenotypic responses such as glucocorticoid-mediated stress responses (Taff et al. 2024).

Variance in biological responses was not affected by thermal fluctuations. This result is commensurate with the finding that short-term fluctuations have a limited effect on phenotypes so that there is little variation between individual traits in their response to fluctuations. The only exception was decreased variation in developmental time in aquatic organisms. A decrease in developmental time may reflect greater energy efficiency and investment into growth by embryos in challenging environments (Pettersen et al. 2023; Vangansbeke et al. 2015) and could also lead to decreased variance.

The shape of the temperature fluctuations can make a difference, and subset analyses for aquatic organisms showed that an alternating fluctuating thermal regime led to an increase in trait values compared to constant temperatures. Diel patterns in natural environments resemble most closely a sinusoidal regime (Bowden and Paitz 2018; Marshall and Burgess 2015; Spanoudis et al. 2015). It may be that the rate of environmental change is important in inducing phenotypic responses (Loughland et al. 2022), and rates may differ between the different thermal regimes reported in the literature. If that were the case, it would have implications for interpretations and extrapolations of laboratory results, and for predicting the impacts of environmental change. For example, acclimation is induced by environmental cues and can remodel phenotypes to mitigate the immediate impacts of environmental stresses (Loughland, Little, and Seebacher 2021; Schulte 2014; Seebacher, White, and Franklin 2015). However, these reversible phenotypic adjustments may depend on the timing between receiving the thermal signal, the rate of thermal change, and initiation of a response (Gabriel 2006; Loughland et al. 2022). Short periods of temperature change can be insufficient therefore to induce acclimation but nonetheless cause cellular damage (Gabriel 2006; Loughland et al. 2022).

Our synthesis of the literature shows that biological responses of ectotherms to thermal change are driven mostly by long-term mean temperatures, and only in certain contexts by temperature fluctuations particularly those with larger amplitudes. Human activity is increasing both temperature means, and the frequency of extreme climatic events (Lee 2023). We identified both as key mediators of the long-term responses of ectotherms. These findings provide crucial insight into the temporal relationship between phenotypic outcomes and fluctuating environments, with the potential to inform conservation approaches: mean temperatures and extreme events such as heat waves (Bauerfeind and Fischer 2014; Breitenbach et al. 2020) are important to consider, but daily temperature fluctuations are of lesser importance. These conclusions pertain to the comparative effects of stable and fluctuating temperatures on single traits. However, temperature fluctuations can modify biotic interactions (Stoks et al. 2017), and environmental temperature profiles can interact with other anthropogenic drivers such as pollution (Chang et al. 2022). Temperature fluctuations can also impact interactions between species because of differential responses to temperature variation. For example, unpredictable temperature fluctuations increased the susceptibility of frogs to pathogens (Raffel et al. 2013), and temperature and soil moisture fluctuations together increased disease susceptibility in a newt (Raffel et al. 2015). Future research should focus on these more complex effects of temperature fluctuations (Verheyen, Delnat, and Stoks 2019).

Author Contributions

Conceptualisation: C.W.S., S.M.B., F.S. Methodology: C.W.S., D.W.A.N. Formal Analysis: C.W.S., D.W.A.N. Investigation: C.W.S., S.M.B., M.J., G.P.F.M., A.K.P., D.R., A.M.R., F.S. Data curation and visualisation: C.W.S. Writing – original draft: C.W.S., F.S. Writing – review and editing: C.W.S., S.M.B., M.J., G.P.F.M., A.K.P., D.R., A.M.R., D.W.A.N., F.S. Supervision: D.W.A.N., F.S. All authors gave final approval for publication.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

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Peer Review

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.